

CURRICULUM VITAE

Calum A Hall

BSc, Computer Science, Undergraduate Student
University of St Andrews

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EDUCATION

University of St Andrews

- BSc Computer Science
- Predicted grade 1:1
- Relevant Coursework: , Computer Security, Critical Systems Engineering, Networking, Databases, Systems Programming, Artificial Intelligence, Data Structures and Algorithms, Human Computer Interaction, Object Oriented Programming

Languages: Java, Python, C, Dart, SQL

Frameworks & Technologies: Flutter, Flask, Jinja, SQLite, Node.js, Git, Express, Junit, Numpy, Pandas, Matplotlib, Scikit-learn, Jupyter Notebooks

George Watson's College

- Advanced Highers in Mathematics, Computer Science, Physics
- Highers in Mathematics, English, Physics, Computer Science, Engineering Science
- Elected Head Boy by peers

Work Experience

Lloyds Banking Group: Worked in the Threat Detection team gaining experience in identifying and responding to potential threats using tools such as Microsoft Sentinel and Splunk. Utilised Kusto Query Language to craft and maintain sophisticated detection rules that monitored network activity for signs of malicious behaviour. These detections were critical in generating actionable alerts, which were then escalated to the Cybersecurity Operations Center.

PROJECTS

Data Visualisation: Worked as part of a five person team building a web page providing a tree visualisation for dinosaur evolutions based on data from Wikidata. My responsibilities included the backend of the site for which I developed a data pipeline that queried the Wikidata API to download data to a JSON file which was then uploaded to a Neo4J database which then served processed data to the front-end using Flask. This process was automated to run every 24 hours.

Scottish Census Data Analysis: Part of a group project that analysed a sample of data from the Scottish census (63,388 individuals) using pandas. Implemented automated consistency checks for the data as some rows were incomplete. Utilised Matplotlib to create several data visualisations and also produced interactive plots using ipywidgets. Created a reproducible Jupyter notebook. Git used for version control.

Film Rating Website: Developed a web application using JavaScript (Node.js), HTML and CSS with the Express framework to browse and manipulate entries in a movie

dataset. Implemented CRUD operations and a JSON-based communication protocol for client-server interaction.

OTHER EXPERIENCE

Oval Analytics Data Collection: Collected data from professional rugby matches and collaborated with data analysts to provide insights and statistics for broadcasters and teams.

Summer Camp Counsellor: Worked at a residential camp in North Carolina for two months, looking after a group of 14-year-old boys.

Oxfam Charity Shop: Volunteered at a local charity shop. Responsibilities included cashier duties, inventory management and customer service.

INTERESTS

Google Student Developer Club: Worked as part of a team to arrange and host technical workshops and talks from industry members and academics.

Music: Achieved grade 8 standard in Guitar and French Horn.

Rugby: Played on school and university rugby teams, captaining sides on a number of occasions.